

BHOOMIKA GUPTA

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EDUCATION

MANIPAL UNIVERSITY JAIPUR B.Tech in Computer Science (AI-ML) - CGPA: 9.91/10 (up to 5 th semester).	August 2023 - Present
JAYSHREE PERIWAL HIGH SCHOOL Senior Secondary (Class XII) - 90.8%	2020-2022
INDIA INTERNATIONAL SCHOOL Secondary (Class X) - 95.6%	2019-2020

SKILLS

- Programming languages: C, C++, Python, SQL, HTML, CSS, JavaScript
- Libraries and Frameworks: NumPy, Pandas, Matplotlib, OpenCV, FastAPI, FAISS
- Core CS Concepts: DSA, Design and Analysis of Algorithms (DAA), Object-Oriented Programming (OOP)
- AI/ML Concepts: Machine Learning, Natural Language Processing, Retrieval-Augmented Generation (RAG), TF-IDF, Cosine Similarity, Web Scraping, Data Preprocessing
- Tools: VS Code, Replit

WORK EXPERIENCE

AI-ML Intern <i>Data Ingenious Global Ltd., Jaipur</i>	May 2025 - July 2025
- Built a content-based movie recommendation system on web-scraped IMDb data, delivering a production-ready solution with an F1-score of 0.78. - Developed a real-time face recognition pipeline integrated with live webcam input achieving an accuracy of ~90%.	

RESEARCH PUBLICATION

- Machine Learning-Based Classification and Repetition Counting of Barbell Exercises Using Wearable Sensor Data** — Springer Nature | [Springer Link](#) **May 2026**
 - Developed a machine learning-based framework for barbell exercise classification and real-time repetition counting using wearable sensor data from accelerometers and gyroscopes.
 - Trained ML models including Random Forest, SVM, Decision Trees, and KNN, achieving up to 98.8% exercise classification accuracy and sub-2% repetition counting error, enabling real-time, scalable workout monitoring.
- Intelligent Pavement Management: A Machine Learning Framework for Condition Classification and Maintenance Prioritization** — Scientific Reports (Nature) | [DOI](#) **April 2026**
 - Benchmarked 8 supervised ML models (Random Forest, XGBoost, SVM, ResNet, and more) for pavement condition classification across urban and rural road networks.
 - Translated model outputs into a five-tier maintenance prioritization framework for actionable road management decisions.

PROJECTS

- Deepfake Detection System** **August 2025**
 - Built a multi-modal detection system combining RoBERTa (text) and EfficientNet-B0 (image/video) for misinformation detection.
 - Implemented Grad-CAM for model explainability and a rule-based module for content risk scoring.
 - Tech Stack: Python, RoBERTa, EfficientNet-B0, Grad-CAM, PyTorch
- Face Recognition System** | [GitHub](#) **July 2025**
 - Designed FaceNet-based embedding generation pipeline for identity matching, with Qdrant enabling scalable fast retrieval, achieving ~90% accuracy.
 - Integrated Qdrant vector database for fast similarity-based retrieval and OpenCV for live webcam detection.
 - Tech Stack: Python, FaceNet, Qdrant, OpenCV, NumPy
- Movie Recommender System** | [GitHub](#) **June 2025**
 - Engineered TF-IDF vectorization and cosine similarity to match movies by genre and plot across the IMDb Top 250 dataset.
 - Evaluated model performance using F1-score, achieving 0.78 for balanced recommendation quality.
 - Tech Stack: Python, TF-IDF, Scikit-learn, Pandas, Web Scraping

ACHIEVEMENTS

- HackSky'25** (August 2025): Secured Top 15 out of 500+ teams at HackSky'25, MIT Bangalore (x Kaspersky), by building an AI platform for real-time misinformation and deepfake detection.
- Awards:** 5x Dean's List Holder every semester for academic excellence (CGPA: 9.91)
- Certifications:** Design and Analysis of Algorithms, Introduction to Machine Learning ([view all](#))